159.102 Instructions for Assignment 1

Assignment 1 starts in Week 2 and is due in Week 5 (**18 August 2022** at 7pm).

You should plan to complete the assignment about a week early (round about 11 August). Then you still have the week to 18 August if something unexpected happens to slow you down.

It is a good idea to put your name and ID number in a comment at the top of your program.

Write a C++ program that converts binary numbers to decimal and decimal numbers to binary. The program prompts the user to enter a number. You must read the number as a string. If the number is binary, convert it to decimal. If the number is decimal, convert it to binary. If the number is not valid, then an appropriate error message must be displayed. Use the examples below to (a) see what is required and (b) see the type of output that can occur.

Notes:

1. Decimal numbers must be in the range 0 to 255 (you must check for this)
2. Binary numbers must not be more than 9 binary digits (you must check for this)
3. A binary number must ***always*** be entered with a leading zero (the first digit you enter must be 0)
4. A decimal number must ***never*** be entered with a leading zero (check for this)
5. Decimal numbers must never be displayed with leading zeros
6. Binary numbers must always be displayed as 8 bits with a space after 4 bits (see the examples)
7. The space will not be used when entering binary numbers – you do not have to check for this
8. There are several different error messages – see examples 4 to 7
9. Do not use a library function to convert binary to decimal – **write your own function**
10. Do not use a library function to convert decimal to binary – **write your own function**
11. You must include at **least two (2) functions** (these could be 9 and 10 above)

Your display must look the same as the examples shown below.

Example 1:

```
Enter a number: 8
Converting decimal to binary. The result is 0000 1000
```

Example 2:

```
Enter a number: 010
Converting binary to decimal. The result is 2
```

Example 3:

```
Enter a number: 10
Converting decimal to binary. The result is 0000 1010
```

Example 4:

```
Enter a number: hello
This is not a valid number.
```

Example 5:

```
Enter a number: 027
This is not a valid binary number.
```

Example 6:

```
Enter a number: 256
This decimal number is outside the range 0 to 255.
```

Example 7:

```
Enter a number: 00101001010
This binary number has more than 9 binary digits.
```

Some general notes about the assignments in 159.102

- You can find the assignment instructions in a file under Assessments and also the start week.
- You submit your assignments via Stream (under Assessments) before the due date and time
- The due date and time appear on the Assignment under Assessments (where you submit)
- **Submit only your .cpp file**
- Do not submit the .exe file or any data files or screen shots of the program running
- Staff are not available to check your assignment before you submit it.
- Do not rush into submitting an assignment. You may find useful information in the notes during the week after the assignment starts.
- Assignments may use C++ knowledge from 159.101, 159.102 and elsewhere. However, if you use knowledge from elsewhere, make sure you use it correctly.

IMPORTANT rules for assignments in 159.102

- You may get assistance when writing an assignment. Assistance includes asking questions and getting ideas from teaching staff and other students. Assistance also includes asking for help when your program is not working correctly and you cannot find the error.
- You may **NOT** get someone else to write your assignment for you. If you submit a program written by someone else, you will lose a significant amount of the marks for the assignment.
- You may **NOT** copy a program from the internet. If you submit a program copied from the internet you will receive **ZERO** marks for the assignment. It is very easy for markers to find the same program on the internet.
- The important thing is that you must show that you understand what is happening in the program you submit. Teaching staff will sometimes arrange zoom sessions with students to check that they understand their submission. If this happens to you, please do not be offended – it is something we have to do as part of the quality assurance for the course.

Working on your assignments in 159.102

- You need an editor/compiler to create and run your program. Atom is provided (see notes to install Atom under Week 1) but you can use any other IDE that supports C++
- Build up your program, for example: start by only converting decimal to binary. When this is working include binary to decimal. Then build in the error checking.
- Give yourself plenty of time. Do not start 6 hours before the deadline!
- Do not give up just because the deadline arrives. You will still get some marks for a partial solution. In a difficult situation, you can apply for an extension.

Marking criteria for assignments in 159.102

Assignments are marked out of 10 and marks can be lost for:

- programs not compiling or running
- errors in code
- programs that have not been tested for a variety of situations
- programs that do not follow the instructions that are provided
- programs that appear to be written by someone else
- programs that are copied from the internet